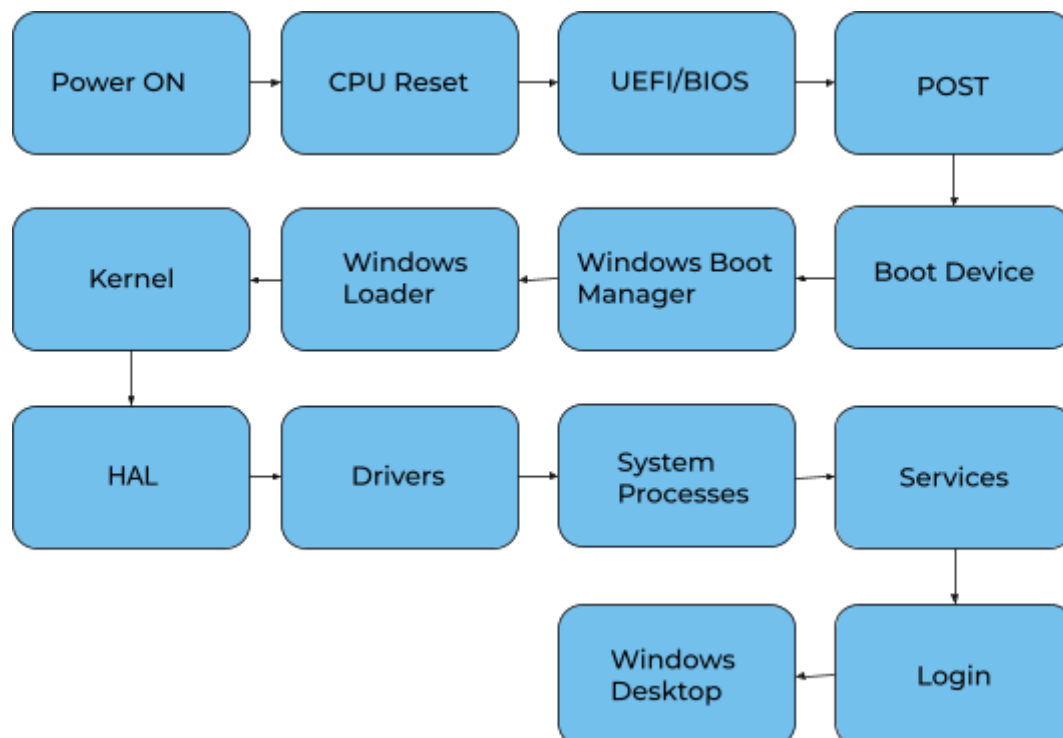


Windows Booting Process

From Power Button to Desktop: How Windows Starts

Introduction

When you start a computer, it uses a built-in mechanism to load the operating system from secondary storage into main memory (RAM). This process is essential because the OS must be present in RAM to manage hardware and software operations. This entire process of loading the OS into memory is called booting. In other words, the boot process is the sequence of operations that begins when you press the power button and ends when the desktop becomes ready. The booting process is necessary to initialize the computer system, load the operating system into memory, conduct hardware checks, ensure system updates for security and stability, and prepare the environment for application execution. The Windows booting process has many steps from power-ON to the Windows desktop, which we will discuss in detail as we further move on.



Types of Booting

There are two types of booting depending on the number of operating systems installed on the machine/computer.

- **Cold or Hard Booting** – Cold booting happens when you turn on your computer after it has been completely turned off. When you press the power button, the computer goes through a series of checks to make sure everything is working properly. This is called a power-on self-test (POST). After this check, it gets the hardware ready and loads the operating system from storage into RAM. This process ensures that your computer is fully set up and ready to use.
- **Soft or Warm Booting** – When you think about "soft or warm booting," it's like gently waking up your computer instead of starting it from scratch. Instead of turning everything off and on again, a soft or warm boot lets the system restart while keeping some of your settings and open programs intact. It's a smoother approach, allowing for a quicker return to what you were doing without losing anything important.

Comparision	Cold Booting.	Warm Booting
Initialized by	Power button.	By pressing Ctrl+Alt+Del.
Performed	Frequently.	Not very common.
Alternate names	Hard Booting, Cold start, and dead start.	Soft Booting.
POST	Included.	Not included.
Basic	Turning a computer ON from a powerless state.	Resetting a computer from an already running state.
Consequence	Does not affect the data or other hardware	Can severely affect the system, causing data loss.

